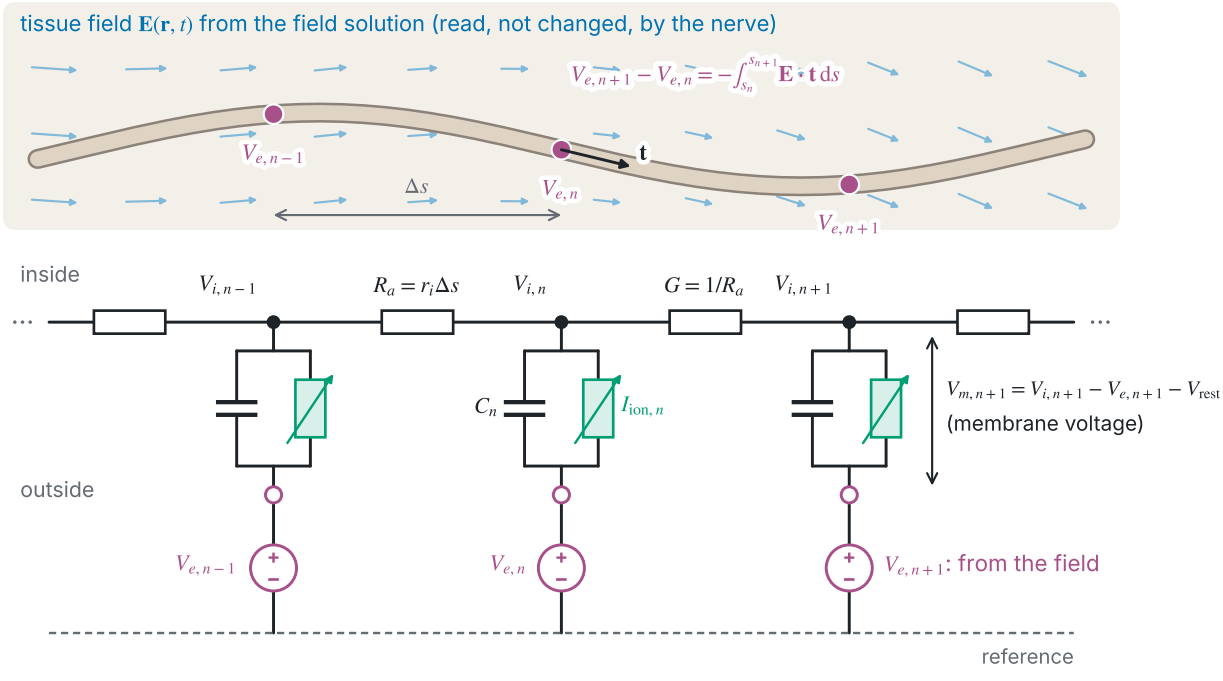
tissue field $\mathbf{E}(\mathbf{r}, t)$ from the field solution (read, not changed, by the nerve)
$V_{e,n+1} - V_{e,n} = -\int_{s_n}^{s_{n+1}} \mathbf{E} \cdot \mathbf{t}\, ds$
$V_{e,n-1}$
$\mathbf{t}$
Δs
$V_{e,n}$
$V_{e,n+1}$
inside
$V_{i,n-1}$
$R_a = r_i \Delta s$
$V_{i,n}$
$G = 1/R_a$
$V_{i,n+1}$
C_n
$I_{\text{ion},n}$
$V_{m,n+1} = V_{i,n+1} - V_{e,n+1} - V_{\text{rest}}$
(membrane voltage)
outside
$V_{e,n-1}$
$V_{e,n}$
$V_{e,n+1}$: from the field
reference